AGENTIC VERTICAL AI

Vision Voice Agent

A repair copilot with eyes and a voice.

It watches what you see, grounds every claim in a pixel, and coaches you step by step — so the skill is on tap the moment you need it.

Try the live demo →

Remember the scene in Matrix where Neo plugs a cable into the back of his head, wakes up a while later, and says "I know Kung Fu." Physical work is about to get something like that — not through brain implants, but through an AI that watches, reasons, and guides.

— the premise, in one paragraph.

WHY NOW

Three forces just converged.

Field work has been untouchable by software because it needed hands, eyes, and judgment in the real world. That gap closed, all at once.

01

Models can finally see.

Multi-modal models now look at a real-world frame and reason about it reliably — objects, state, what just changed. That capability didn't exist eighteen months ago.

02

The hardware is already here.

Phones, AirPods, smart glasses — a camera, a mic, and a speaker in the worker's sightline. No new device needs to ship.

03

Skilled labor is scarce.

Field services, manufacturing, nursing, HVAC — every trade is short people. Training takes months to years. Time-to-skill is the whole game.

A grounded observer. A natural voice.

The hard part isn't talking — it's not hallucinating. So the agent has two brains: one writes down only what's literally in the frame; the other speaks like a friend watching over your shoulder.

STEP 01 · PERCEPTION

Cold observer writes the facts.

Gemini 2.5 Flash returns strict JSON per frame — objects, device_state, visible_features, changed_vs_prior. Ground truth only, no user-action inference.

STEP 02 · GROUNDING

Voice model narrates against those facts.

Gemini Live sees the same frame with its own eyes but must use the JSON as a guardrail. It can't contradict it. Natural delivery, no hallucinations.

STEP 03 · COACHING

Manuals and web search on demand.

Tools load mid-conversation — screw counts, hidden fasteners, model-specific warnings. The agent doesn't advance until the camera confirms the step.

FRAME JPEG every 2s · 640×480 · base64 · WS

↓

DIFF PIL 32×32 thumbnail · threshold 12

↓

SEE Gemini 2.5 Flash · strict JSON · cold observer

↓

GUARD observation_signature dedup · anti-hallucination

↓

TOOLS lookup_manual · web_search · on demand

↓

SPEAK Gemini Live · native audio · own eyes + JSON

TOOLS

It calls tools when the answer lives outside its head.

The agent isn't just a camera and a voice. When a brand name appears or a model number is asked, it reaches for the right source — without breaking flow.

lookup_manual

INTERNAL · RAG

A library of repair playbooks indexed per device. Pulls exact procedures — screws, tools, hidden fasteners — and narrates them step by step.

CALL

```
tool_call(lookup_manual)
  query: "Morphy Richards trimmer disassembly"
```

RESULT

```
total_screws: 3  # 1 back + 2 hidden
tool: "H1 hex"
warnings: ["battery still live"]
```

AGENT SPEAKS

```
"Three H1 screws – one at the back,
 two hidden under the front hood.
 Battery stays live, don't short it."
```

web_search

LIVE · GOOGLE

When the manual doesn't cover it — rare models, error codes, recalls — the agent searches the live web, summarizes, and speaks the fix.

CALL

```
tool_call(web_search)
  query: "DeLonghi EC155 E3 error reset"
```

RESULT

```
source: "delonghi.com/support"
summary: "E3 = pump priming fault.
 Hold STEAM + COFFEE 5 sec."
```

AGENT SPEAKS

```
"E3 means the pump didn't prime.
 Hold STEAM + COFFEE five seconds,
 then run a water cycle."
```

OFFLINE MODE

Breakdown in a dead zone? It still works.

Car stalls on a no-signal stretch of highway. Factory's air-gapped. Offshore rig. Hospital that blocks outbound traffic. The two-stage architecture is model-agnostic — swap the cloud backbone for on-device models and zero bytes leave the device.

Cloud mode

DEFAULT

Best voice, richest reasoning, live tool calls. Default path when the network is strong.

```
STACK
voice:       Gemini Live · native audio
perception:  Gemini 2.5 Flash · JSON
tools:       lookup_manual · web_search
latency:     ~300 ms to first word
```

On-device fallback

OFFLINE · 0 BYTES

A laptop, phone, or Pi-class board runs the whole loop locally. Same prompts, same tool schemas, same UI — only the model backbone swaps.

```
STACK
voice in:    whisper.cpp · ggml
perception:  Moondream / Qwen2-VL
voice out:   Piper · neural TTS
manuals:     local vector store
```

WHERE IT SCALES

One agent. Every trade that works with its hands.

Field services

Installers and technicians get a senior engineer in their ear — first visit looks like the hundredth.

HVAC & appliances

Exact screw counts, tool sizes, and hidden fasteners called out in real time.

Manufacturing & QA

Visual checks catch errors before the part moves downstream.

Nursing & care

Protocols narrated and verified as a glove goes on or a dose is drawn.

THREE WAYS TO BUILD IT

The technology is the easy part. Distribution is the question.

APPROACH 01

Sell into existing workforces.

Ship to companies that already employ field teams. Immediate productivity; distribution through enterprise contracts.

APPROACH 02

Own a vertical end-to-end.

Pick one trade — HVAC, appliance repair, nursing — and build a super-powered workforce. The AI is why our workers outperform.

APPROACH 03

A platform anyone can join.

Sign up, pick up the skill, start earning — with the agent as instructor, supervisor, and quality bar.

Give physical workers AI superpowers.

If you run a field team, a repair business, or a training program — we'd like to show you what the agent does with a live camera and a real job.

Try the live demo →

Read the landing page →